

SIEMENS CIOS SELECT QUICK MANUAL



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www.siemens.com/healthineers

Cios Select

Quick Guide



Manufacturer's notes:

This device bears a CE mark in accordance with the provisions of EU Regulation 2017/745 of April 5, 2017 concerning medical devices and the Council Directive 2011/65/EU of June 8, 2011 on the restriction of the use of certain hazardous substances in electrical and electronic equipment.

The CE marking applies only to Medical Devices which have been put on the market according to the above-mentioned EU Regulation and EU Directive.

Unauthorized changes to this product are not covered by the CE mark and the related Declaration of Conformity.

Original language: English

Introduction

We welcome you as a user of the Cios Select mobile X-ray system from Siemens.

This Quick Guide is a guideline to help you with the operation of the system.



The Quick Guide is valid only in conjunction with the operator manuals and the safety information they contain:

- ☐ Please observe the operator manual and all supplements/addenda
- ☐ Please observe all safety information

The right-hand pages contain step-by-step instructions corresponding to the typical workflow in the intensive care unit.

The pages to the left contain illustrations and supplemental notes.

The description focuses on the simple and quick operation of the Cios Select from starting up to shutting down the unit.

More detailed and complete descriptions can be found in the Cios Select Operator Manual.

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Subtraction/Roadmap

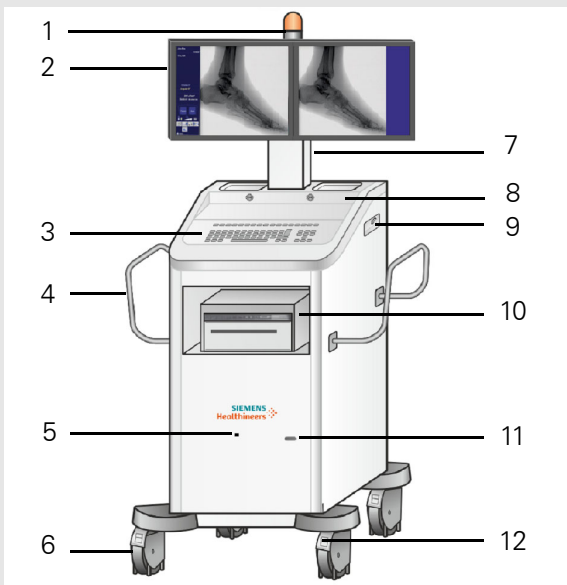
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6 Quick Guide Cios Select



System overview

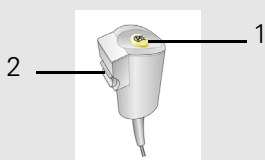
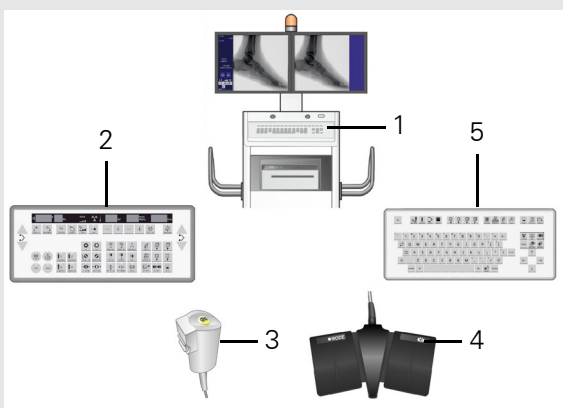
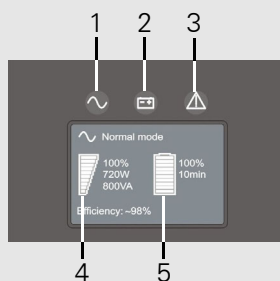
Cios Select C-arm system

1. Horizontal support arm
2. Handles including holder for hand switch, steering handle for transport
3. Connection for central plug
4. Base system (chassis) with electronics unit
5. Single tank with X-ray tube unit and collimator
6. Image intensifier with grid

Monitor trolley

1. Radiation display
2. Monitors
3. Control panel
4. Handles: for neat cable storage
5. Reset key
6. Wheels with cable deflectors
7. Monitor column
8. Storage for mouse; USB connections
9. Connection for WLAN, LAN
10. Shelf for printer
11. UPS window
12. Locking brake

System overview



System overview

UPS window

The UPS window indicates the battery charge level and the system operating status.

1. LED "Green": Input voltage (line power operation)
2. LED "Yellow": Battery operation
3. LED "Red": Alarm, malfunction
4. Bar display of utilization ratio
5. Bar display of battery charge status

Overview of operating elements

1. Control panel on the monitor trolley
2. Control panel on the C-arm
3. Hand switch
4. Standard footswitch
5. Keyboard on monitor trolley

Hand switch

Hand switch: to remotely control radiation release and image storage.

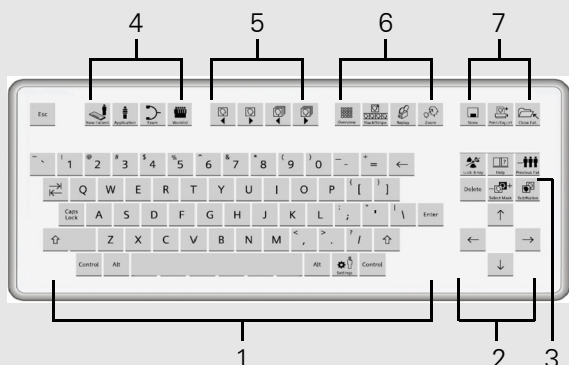
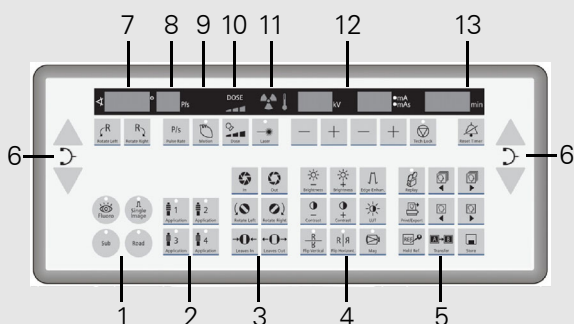
1. Yellow button for radiation release
2. Save button to save images/scenes

Footswitch

Footswitch: used to remotely control radiation release with the following default setting:

1. Release of preset mode
2. Fluoroscopy (Fluoro)

System overview



C-arm keyboard

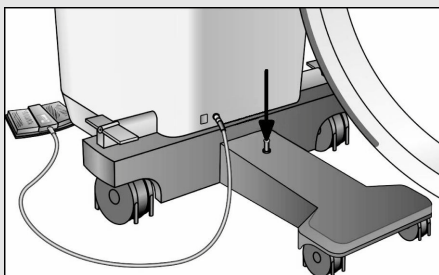
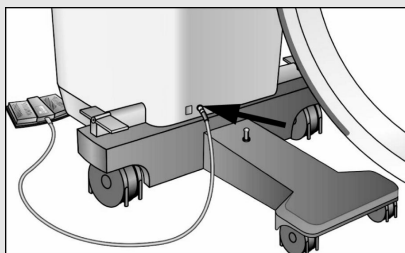
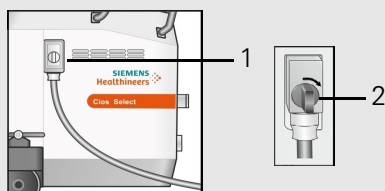
The keyboard for performing your examinations is located on the C-arm.

1. Selection of operating mode
2. Pre-defined application settings
3. Collimator setting
4. Image processing
5. Image display, documentation
6. Lift/lower C-arm
7. Image rotation (left/right)
8. Pulse rate
9. Noise filter selection
10. Select and display dose rate level
11. Radiation indicator, tube load indicator, select laser light localizer
12. Display/manually select radiation parameters
13. Radiation time display, reset radiation time

Keyboard on the monitor trolley

The control panel for preparing and analyzing examinations is located on the monitor trolley.

1. Keyboard area
2. Cursor keypads
3. Special functions (e.g., radiation block, subtraction display)
4. Patient registration and examination
5. Scrolling buttons
6. Image display
7. Documentation, close patient



Preparing Cios Select

Connecting the C-arm system to the monitor trolley

- ❖ Plug the main monitor trolley connection cable plug into the socket on the C-arm basic unit (1).
- ❖ Turn the switch to the right until it locks in (2).

Connecting the footswitch

- ❖ Insert the footswitch cable into the left socket in the connection area on the C-arm basic unit.

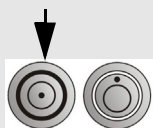
Establishing the equipotential bonding connection

- ❖ Clamp the equipotential bonding cable to the center socket in the connection area of the C-arm basic unit and to an equipotential bonding point in the vicinity of the patient.

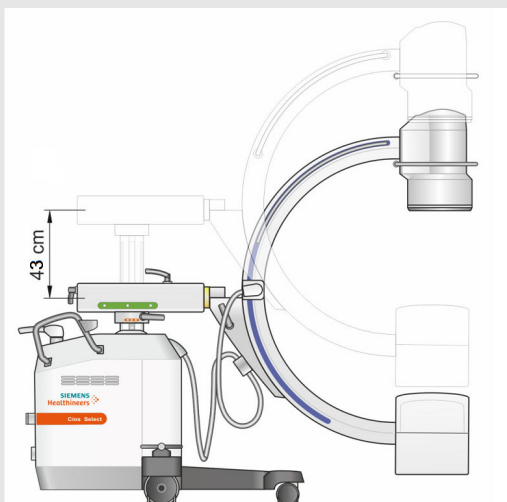
Connecting to power

- ❖ Plug the power plug into the appropriate socket.

Startup



- ☐ An audible signal sounds while the system starts up.
- ☐ Do not press any button on the control panel during startup, otherwise an error message will appear.



Switch on Cios Select

- ❖ Press the right **ON** button on the C-arm basic unit.
- The C-arm system starts up.
- An acoustic signal sounds during startup.

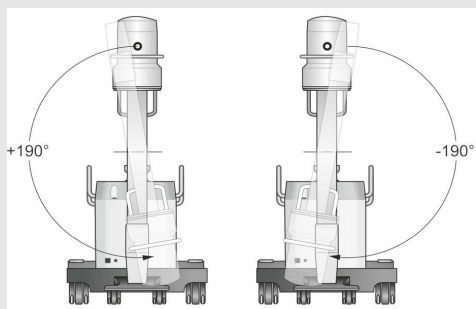
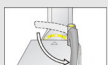
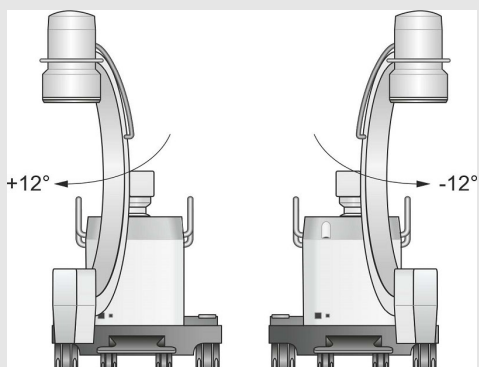
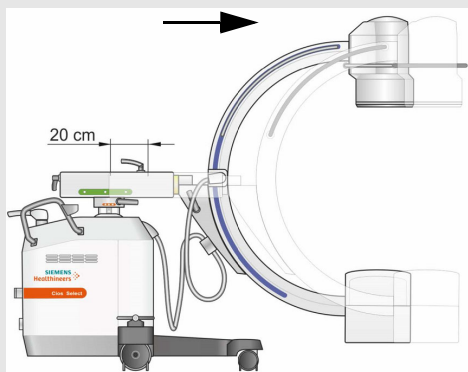
Wait for the imaging system to boot.

Lifting and lowering the C-arm

You can lift and lower the C-arm by motor control using the arrow buttons on the C-arm basic unit.

Moving the C-arm vertically

- ❖ Press the Up or Down button on the C-arm system control unit.
- The lifting column remains in an interim position during lowering and then lowers completely when the button is pressed again. An audible signal sounds.



Moving the C-arm horizontally

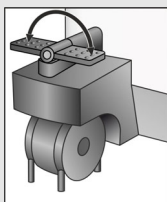
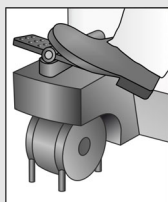
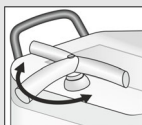
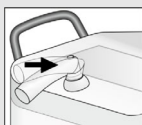
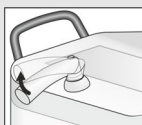
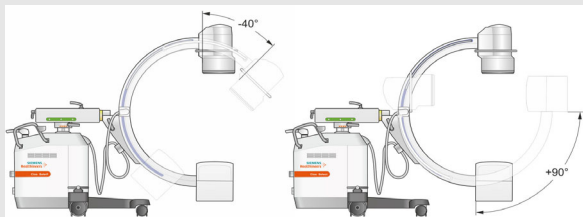
- ❖ Open the brake lever marked in green for *horizontal movement* by turning it to position.
- ❖ Move the support arm while observing the green scale.
- ❖ Re-engage the brake.

Swiveling the C-arm

- ❖ Open the brake lever marked in orange for *horizontal swivel* by turning it to position.
- ❖ Swivel the C-arm.
- ❖ Re-engage the brake.

Angulating the C-arm

- ❖ Open the brake lever marked in yellow for *angulation* by turning it to position.
- ❖ Rotate the C-arm while observing the yellow scale on the support arm joint.
- ❖ Re-engage the brake.



Orbital movement of the C-arm

- ❖ Open the brake lever marked in blue for *orbital movement* by turning it to position.
- ❖ Rotate the C-arm while observing the blue scale.
- ❖ Re-engage the brake.

Transport of the Cios Select

Transport the Cios Select as follows:

1. Drive operation and steering of the C-arm system

- ❖ Pull the lever up.
- ❖ Lift the steering lever and turn it in the desired direction.
- ☐ Motion straight ahead / pivoting in place
- ☐ Diagonal motion
- ☐ Straight left or straight right motion

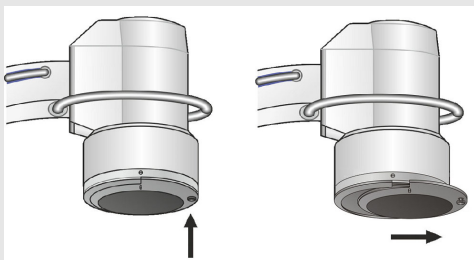
The Cios Select can now be moved in the desired direction.

2. Braking the C-arm system

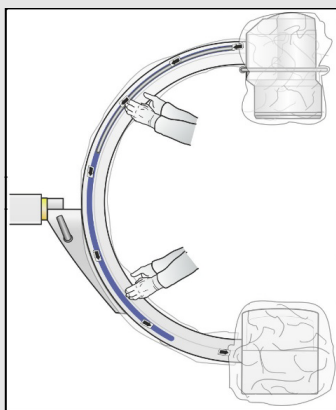
For the desired park position, engage the brakes as follows:

- ❖ Keep the foot pedal pressed on either side of the basic unit. The foot pedal can be pressed in both directions.
- The basic unit brakes are now locked.

Examination procedure



These icons on the left monitor and the control panels indicate the “inserted/removed” status of the scatter radiation grid.



Removing the scatter radiation grid

If necessary, remove the scatter radiation grid as follows:

- ❖ Unlock the scatter radiation grid by turning the attachment screw.
 - ❖ Carefully pull the scatter radiation grid out and set it safely aside.
 - ❖ Check that the icon for “scatter radiation grid inserted” has a line through it.
- To insert the scatter radiation grid, follow the directions above in the reverse order. Then the “scatter radiation grid inserted” icon must appear.

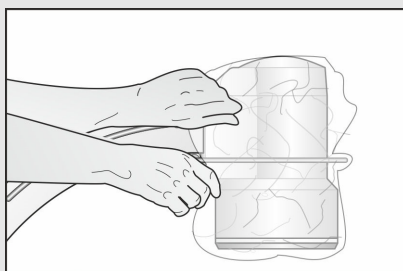
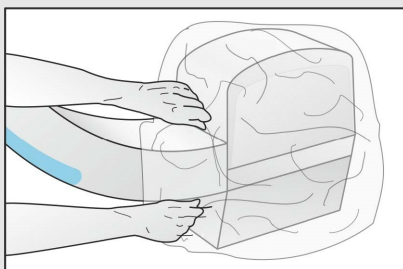
Attaching the sterile covers

C-arm

To protect it against contamination, the C-arm including the image intensifier and single tank are completely covered with a three-piece sterile disposable cover of transparent polythene sheet.

- ❖ Attach the sterile cover to the inside of the C-arm using the adhesive strips.

Examination procedure



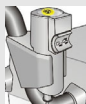
Examination procedure

- ❖ Pull the plastic cover over the single tank.
- The plastic cover is fixed in place with an elastic cord.

- ❖ Pull the other plastic cover over the image intensifier.
- The plastic cover is fixed in place with an elastic cord.

The C-arm is now covered completely.

Examination procedure



Emergency



[New patient](#)

Data Entry Dialog

PATIENT Last Name First Name Patient ID v Date of Birth D M Y Age Sex ☐ Male ☐ Female ☐ Other	INSTITUTION Operator 1 Operator 2 Referring Physician Physician 1 Application Application group Application
STUDY Accession No. Request ID Study Description Study ID Study Comment	



Preregister

Examination procedure

Patient registration

Emergency patient

- ❖ As long as no other patient is registered, press the hand or footswitch once for radiation release.

Or

- ❖ Click the **Emergency** button in the **Data Entry Dialog** window.
- The patient is registered using temporary patient data that can be corrected later.

New patient

- ❖ Press **New Patient** button on the control panel.
 - ❖ Enter the patient data in the data entry dialog on the left monitor.
 - Name, Patient ID, Age, and Sex of the patient are always mandatory entry fields (in bold). This information uniquely identifies the patient in your databases.
 - ❖ If necessary, you can enter a unique patient ID using the **Generate ID** button.
 - ❖ Press the **Exam** button to examine the patient immediately.
- Or
- ❖ Click the **Preregister** button in the **Data Entry Dialog** window to flag the patient for examination at a later time.

Examination procedure



▲ Patient Name	Patient ID	Date of Birth	Sex	Study Description	Physician
DSA_Kidney Female_1942	0060971	01-01-1942 (71)	F	DSA_General	
Elbow child pre and pos...	0005125617	29-11-2004 (8)	F	Rö-Ellembogen re...	
Emergency 477	E_477	22-05-2013 (0)	O	Standard	
Emergency 477	E_477	22-05-2013 (0)	O	Standard	
Emergency 477	E_477	22-05-2013 (0)	O	Standard	
Emergency 478	E_478	22-05-2013 (0)	O	Standard	
Emergency 479	E_479	22-05-2013 (0)	O	Standard	
Hand right	500651758	18-02-1988 (25)	M		
Hand right	500651758	18-02-1988 (25)	M	Standard	
Hand right	500651758	18-02-1988 (25)	M	Standard	



Examination procedure

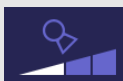
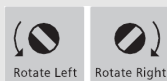
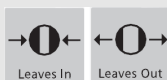
Previous or pre-registered patient

- ❖ Press the **New Patient** button on the control panel and press one of the following buttons:
 - ☐ **Previous patients** if the patient was examined once previously.
 - ☐ **Worklist** if the patient was flagged for an examination via the HIS/RIS or using the data entry dialog.
- ❖ Use the arrow keys to select the patient from the list.
- ❖ Double-click the patient with the mouse.
- The selected entry is copied into the Data Entry Dialog.
- ❖ Press **Exam**.

Examination procedure



Adjust the exposure parameters on the left side of the control panel.



Defining the exposure parameters

Operating mode

- ❖ Press the button for the desired mode.
- The button shaded in white indicates the current mode.

Collimators

- ❖ Open or close the iris collimator with the **In** or **Out** buttons.
- ❖ Open or close the semi-transparent slot collimator with the **Leaves In** or **Leaves Out** buttons.
- ❖ Rotate the semi-transparent slot collimator to the left or right with the **Rotate Left** or **Rotate Right** buttons.

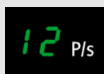
Dose rate level

- ❖ Press the **Dose** button.
- ❖ Use the appropriate button multiple times to set the dose to a preferred setting of low, medium or high.
- The bar indicator on the left monitor and control panel shows the currently set dose level.

Examination procedure



k factor: a number of k exposures are integrated into one image (depending on mode and examination program, from $k = 1$ to $k = 16$).



Examination procedure

Motion/noise filter

- ❖ Press the **Motion** button.
- Turn off/on motion detection for fixed or moving objects.

Selecting the input format (I.I. zoom/magnification)

- ❖ Press the **Magnify** button.
- Each time you press the key, you switch between zoom levels 0 (full format) and 1.

Zoom level	Input format
Mag 0	23 cm (full format)
Mag 1	16 cm

Pulse rate

- ❖ Press the **Pulse rate** button multiple times if needed.

→ The pulse rate changes accordingly.

The k value currently set is shown on the display of the C-arm keyboard above the Pulse Rate button.

If continuous is set, "co" (continuous) appears.

Examination procedure



Tech lock is recommended, for example

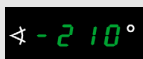
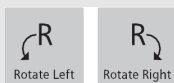
- ☐ When metal parts come into the beam path (due to rods from surgeries, for example)
- ☐ Fluoroscopy of objects with widely varying densities (such as prosthetic hips)



Tech Lock



Image parameters are entered separately on the C-arm keyboard. Changes affect the current and subsequently captured images.



Examination procedure

Tech lock – manual input of radiation parameters

Automatic dose regulation (ADR) is usually used.

Deactivating the button turns off this control. The radiation parameters can then be changed manually.

- ❖ Press the **Tech lock** button.
- Automatic dose regulation stops.
The kV/mA or kV/mAs level remains constant at the most recently set and displayed values.
- ❖ Use the **+/-** to manually set other values, if desired.
- The set values are shown on the control panel.

Setting the image parameters

Image rotation and image mirroring

- ❖ Press the **Rotate Left** or **Rotate Right** button.
- Adjust the angle of rotation in 1° or 90° increments using the image rotation buttons.
- ❖ Select horizontal and/or vertical mirroring using the **Flip Horizont.** and **Flip Vertical** buttons.
- The current image orientation is displayed in the image area of the control panel.

Examination procedure

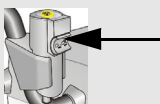
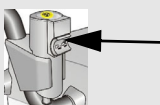
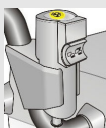
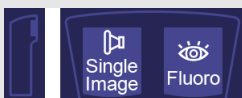


Check the left monitor display to ensure the current patient and the registered patient are the same.

Meier, Hans

23-12-1976

Standard



Acquiring images

All further steps of the examination are performed at the C-arm system.

Release radiation

- ❖ Release radiation with the hand switch or foot-switch.
- The current fluoro image is displayed on the left (live) monitor, the LIH after radiation is complete.

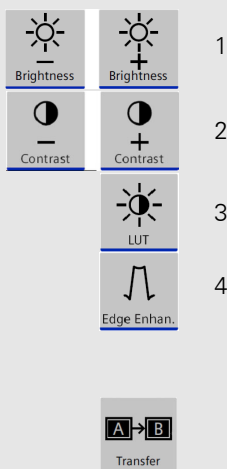
Saving images

- ❖ To save images during or after radiation, press the **Store** button on the control panel or on the hand switch (< 2 seconds).
- The current image is saved in the local database and displayed on the right monitor.

Saving a scene (LSH)

- ❖ To save a scene during or after radiation, press the **Store** button on the control panel or on the hand switch (> 2 seconds).
- The current scene (LSH) is saved in the local database and displayed on the right monitor.

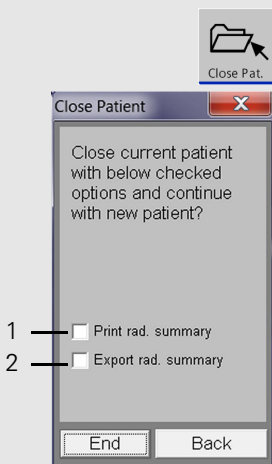
Examination procedure



If the **Transfer** button is pressed, the transferred image is not saved.



Images/scenes are automatically transferred to the right monitor with the save command.



Examination procedure

Changing the image display

- ❖ If needed, use the following tools to improve the image display:
 - ☐ Brightness (1)
 - ☐ Contrast (2)
 - ☐ LUT selection (3)
 - ☐ Edge enhancement (4)

Using reference images

- ❖ Press the **Transfer** button to transfer the live image to the right (reference) monitor.

- ❖ Press the **Hold reference** button to hold the currently displayed reference image.

Ending the examination

- ❖ Press the **Close pat.** button at the lower right of the control panel.
- ❖ Choose the required options:
 - ☐ Print rad. summary (1) on the default printer
 - ☐ Export rad. summary (2) to the default export address

- ❖ Press the **End** button.

Postprocessing



Previous pat.

Patient Name	Patient ID	Date of Birth	Sex	Study Description	Physician
DSA_Kidney Female_1942	0060971	01-01-1942 (71)	F	DSA_General	
Elbow child pre and pos...	0005125617	29-11-2004 (8)	F	R0-Ellembogen re...	
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Emergency 477	E_477	22-05-2013 (0)	O	Standard	
Emergency 478	E_478	22-05-2013 (0)	O	Standard	
Emergency 479	E_479	22-05-2013 (0)	O	Standard	
Hand right	500851758	18-02-1988 (25)	M		
Hand right	500851758	18-02-1988 (25)	M	Standard	
Hand right	500851758	18-02-1988 (25)	M	Standard	

1

2



You can organize the list display as desired using the appropriate filter and by sorting the columns.



▲ Patient Name



Settings



Correct pat. data

Patient data

Displaying the Patient list

- ❖ Press **Previous Patient** button on the monitor trolley keyboard.
- The Patient list displays the data stored in the local database for previously examined patients:
 - ❑ Study list (1), the selected study is marked in blue
 - ❑ Preview images of the selected study (2)

Selecting data

- ❖ On the monitor trolley keyboard, press the first letters of the required patient.
- The first name with a suitable patient name is selected in the **Patient list**.
- ❖ Use the arrow buttons or mouse to select the required study.

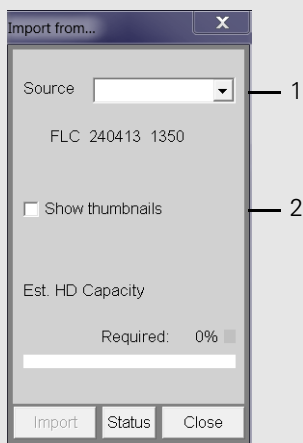
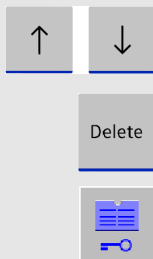
Correcting data

- ❖ Select the data set to be corrected in the patient list and press the **Settings** button.
- The **Manage Tool** is displayed on the left monitor.
- ❖ Click this button in the **Patient** subtask card.
- The data of the selected patient are transferred to the **Data Entry Dialog** window.
- ❖ Confirm the changes by pressing the **Correct pat. data** button.

Postprocessing



You can configure automatic deletion of images based on their processing status.



If the "DICOM node" source is selected, the **Query for retrieve** dialog box is displayed for the data import.

Deleting data

To ensure sufficient storage capacity, you should regularly delete images that are no longer needed from the local database.

- ❖ Select the entry to be deleted in the Patient list.
 - ❖ Press this button.
- If needed, you can activate/remove deletion protection using this button.

Importing data

- ❖ Prepare the appropriate data medium (insert CD/DVD, USB stick).
- ❖ Click the import icon at the bottom left of the left monitor.
- ❖ Enter the import settings:
 - ☐ Data source CD/DVD, USB, DICOM node (1)
 - ☐ Show thumbnails for CD/DVD, USB storage media (2)
- ❖ Select the desired items in the study list of the storage medium displayed.
- ❖ Click **Import**.



★ Patient Name	Patient ID	Date of Birth	Sex	Study Description	Physician
DSA_Kidney Female_1942	0060971	01-01-1942 (71)	F	DSA_General	
Elbow child pre and pos...	0005125617	29-11-2004 (8)	F	R6-Elleboog en...	
Emergency 477	E_477	22-05-2013 (0)	O	Standard	
Emergency 477	E_477	22-05-2013 (0)	O	Standard	
Emergency 477	E_477	22-05-2013 (0)	O	Standard	
Emergency 478	E_478	22-05-2013 (0)	O	Standard	
Emergency 479	E_479	22-05-2013 (0)	O	Standard	
Hand right	500651758	18-02-1988 (25)	M		
Hand right	500651758	18-02-1988 (25)	M	Standard	
Hand right	500651758	18-02-1988 (25)	M	Standard	



Editing images

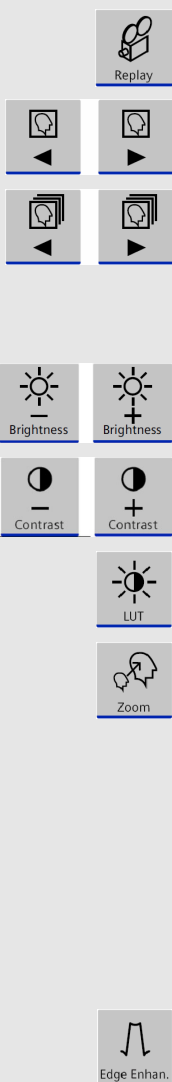
Loading images

- ❖ Press **Previous Pat.** button on the control panel.
- ❖ Use the arrow buttons to select the desired study.
- ❖ Double-click the patient with the mouse.

Displaying images

- ❖ Turn on **Overview** if needed.
- ❖ Select from the following displays of loaded images on the left monitor:
 - ☐ Stack display: One series per image segment.
 - ☐ Stripe display: One image from the selected series per image segment.
 - ☐ Press **Stack/Stripe** to toggle between the display modes.
- Use the **Scroll** buttons to display other series in gallery display.

Postprocessing



Reviewing a scene

- ❖ Press the **Replay** button.
- ❖ Press one of these buttons to go backward/forward by a single image.
- ❖ Press one of these buttons to go to the previous/next scene.

Windowing images

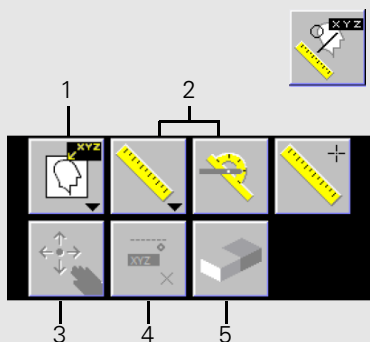
- ❖ Use **Brightness** button to change the brightness.
- ❖ Use **Contrast** button to change the contrast.
- ❖ Use **LUT** buttons to select user-specific window value settings.
- ❖ Use **Zoom** button to change zoom level between 1 (full image) to 1.5, to 2.

Using image filters

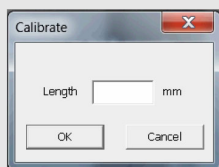
You can postprocess blurred images with filters.

- ❖ Press the **Edge enhancement** button.
- ❖ Select a suitable edge filter to increase the highlighting of structures.

Postprocessing



Analyses using distance measurements require previous calibration of the image.



Evaluating images

Start the evaluation functions

- ❖ Click this button on the left monitor (lower left).

The following tools are available:

- ☐ Annotate images (1)
- ☐ Line and angle measurement (2)
- ☐ Graphic object selection (3)
- ☐ Delete individual graphic object in image (4)
- ☐ Delete all graphic objects in image (5)

Calibration

For calibration, draw a distance line along an object of known dimensions. The object must be positioned vertical to the X-ray beam and the distance line must lie in the same plane.

- ❖ Click this button and draw a distance line along the calibration object.
- ❖ Click this button to calibrate.
- ❖ Enter the known length of the distance line (in mm). Confirm with Enter.

Measuring the distance

- ❖ Click this button.
- ❖ Click the starting point of the line.
- ❖ Click the desired end point.
- The length (in mm) and the sequential number of the graphic object are displayed in the image.



Depending on the start and end points of the second leg, either the interior or exterior angle is indicated.



Text	
L	R
text	
Add to predefs.	Del from predefs.
Comment	
text	
Add to predefs.	Del from predefs.



L
R



The side identification is applied automatically to the remaining images of the series.

Measuring an angle

- ❖ Click this button.
- ❖ Draw the first leg by setting the desired start and end points with the mouse pointer.
- ❖ Draw the second leg using the same method.
- The registered angle (in °) and the sequential number of the graphic object are displayed in the image.

Annotating images

- ❖ Click this button to activate the text drawing mode.
- ❖ Enter the note and confirm using Enter.
- ❖ Move the mouse pointer to the location for the annotation.
- ❖ *As an alternative:* Copy predefined text from an existing template.

Setting the side identification

- ❖ Click this button to activate the text drawing mode.
- ❖ Click the side identification button for the correct patient side.
- ❖ Move the mouse pointer to the location for the side identification.
- The selected identification **L** or **R** is inserted at the position of the mouse pointer.

Postprocessing



All changes in the image are saved when you close the patient.



Manually saving images and scenes

During manual saving, the selected images/scenes are saved as copies of the original in the corresponding study.

Saving single images

- ❖ Use the **Ctrl** and **Shift** keys to select the desired sequential images.
- ❖ Press the **Store** button.

Saving partial scenes

- ❖ Use the **Ctrl** and **Shift** keys to select the desired sequential images.
- ❖ Press the **Store** button.



Printer PaperPrinter 1

Name 2

Size A6/L 2

Layout 1x1 2

SUB/ROAD images

☒ Print as subtracted 3

Copies 1 4

☒ Anonymous
☐ Hide Text 4

5

Print

Selecting images

- ❖ Press the **Overview** button on the monitor trolley keyboard.
- ❖ Select the desired individual images or series by holding down the **Ctrl** button.

Printing using default settings

- ❖ Press the **Print/Export** button.
- The default printer is used with the defined paper size and segmentation settings.

Printing with changed settings

- ❖ Click the print icon at the bottom left of the left monitor.
- ❖ Enter the print settings:
 - ☐ Printer (1)
 - ☐ Layout (2)
Print layout, page size, segmentation
 - ☐ Display subtraction images (3)
 - ☐ Number of printed copies, text display (4)
- ❖ Click **Print** (5).



To export complete studies, select the records in the **Patient list**.



During data transmission, do not shut down the Cios Select, disconnect the monitor trolley from the C-arm system, or remove the data storage medium.



The appearance of the **Print/Export** button depends on the set default export target.

Exporting

Selecting images

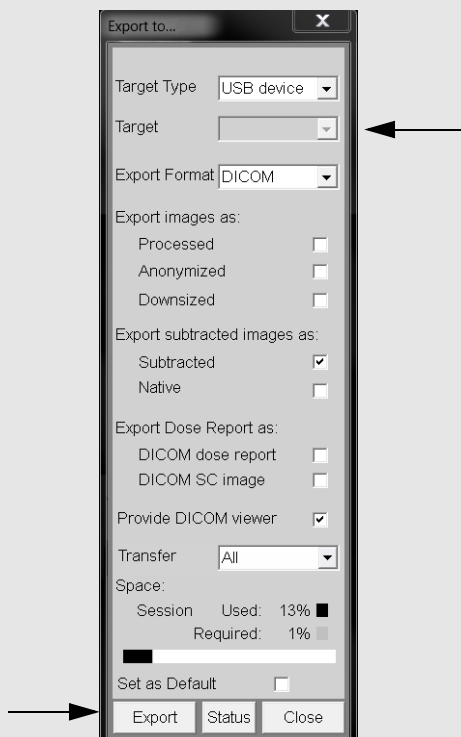
- ❖ Press the **Overview** button on the monitor trolley keyboard.
- ❖ Select the desired individual images or series by holding down the **Ctrl** button.

Preparing the removable devices

- ❖ Place the CD/DVD into a protable drive with the label facing upwards.
- ❖ Connect the USB plug into the USB jack.

Exporting using default settings

- ❖ Press the **Print/Export** button.
- Exports are sent to the default destination using the settings defined.



When this option is activated, the current parameters are set to the default export settings.

Set as Default



Exporting with changed settings

- ❖ Click the export icon at the bottom left of the left monitor.

- ❖ Enter the export settings:
 - ☐ Export destination DICOM node, CD/DVD, USB
For a DICOM node, the network address (**Target**) must be entered
 - ☐ Data format
Graphic, video, DICOM or private (internal to system)
 - ☐ Image properties
with/without graphics or text, reduced file size by scaling the image resolution to 50%

 - ☐ Selection of image types for subtraction images

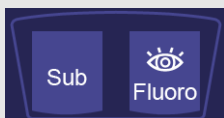
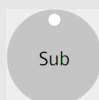
 - ☐ Data format of radiation report
 - ☐ Also save DICOM viewer on data storage medium

- ❖ Check for sufficient space on data storage medium.
- ❖ Click **Export**.

Subtraction/Roadmap



During the examination, images without contrast agent (mask) are continuously subtracted from images with contrast agent and displayed on the monitor. Depending on the contrast agent flow, they display the relevant vascular segment without superimposition in real time.



Performing subtraction angiography

Subtraction angiography is divided into the following three phases:

- ❑ Phase A
Creating the mask (defined, fixed duration)
- ❑ Phase B1
Contrast medium injection (duration of "Inject" display until examination area is reached)
- ❑ Phase B2
Time of the actual exposure of the examination region

Preparation

- ❖ Register the patient.
- ❖ Press the **Sub** button.
- ❖ Enter the necessary settings for the examination.

Phase A

- ❖ Release radiation with the hand switch or the **Sub** pedal of the footswitch.
- During the generation of the mask the native image is displayed on both monitors.

Phase B

- ❖ Keep the exposure release button pressed.
- ❖ Inject the contrast medium when the syringe icon displays (screen and control panel).
- The control panel and the left monitor display the subtraction image.
- The native image on the right monitor displays the blood vessel as it is continuously filled with contrast medium.



The mask (phase A), subtraction series (phase B) and the fill image are saved automatically.

Subtraction/Roadmap

Fill image

- ❖ Do not turn off the radiation until the contrast medium has completely passed through the vessel.
- The monitor displays the maximum fill image (PeakOP image).

Subtraction/Roadmap



▲ Patient Name	Patient ID	Date of Birth	Sex	Study Description	Physician
DSA_Kidney Female_1942	0060971	01-01-1942 (71)	F	DSA_General	
Elbow child pre and pos...	0005125617	29-11-2004 (8)	F	Rö-Ellembogen re...	
Emergency 477	E_477	22-05-2013 (0)	O	Standard	
Emergency 477	E_477	22-05-2013 (0)	O	Standard	
Emergency 477	E_477	22-05-2013 (0)	O	Standard	
Emergency 478	E_478	22-05-2013 (0)	O	Standard	
Emergency 479	E_479	22-05-2013 (0)	O	Standard	
Hand right	500651758	18-02-1988 (25)	M		
Hand right	500651758	18-02-1988 (25)	M	Standard	
Hand right	500651758	18-02-1988 (25)	M	Standard	



Subtraction postprocessing

Load images (after examination is complete)

- ❖ Press the **Previous Pat.** button on the control panel.
- ❖ Use the arrow buttons to select the subtraction study.
- ❖ Double-click the patient with the mouse.

Changing the monitor layout

- ❖ Press the **Sub/Native** button and select display on both monitors:
 - ☐ 2-channel display: Subtraction image (left) and associated native image (right).
 - ☐ 1-channel display: only native images.
 - ☐ 1-channel display: only subtraction images.

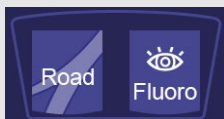
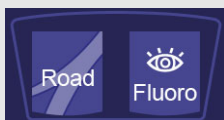
Scrolling/displaying images

- ❖ Switch to 1-channel display.
- ❖ Use the image overview to scroll through the images/series.
- In stack display the PeakOP image of the respective series is displayed.

Subtraction/Roadmap



The fill image can be used from previous subtraction angiography or it can be recreated in a like manner in **Road** mode.



Performing roadmap

In the first step of the Roadmap examination, the fill image is created as a mask from a normal subtraction (phase A and phase B).

In the second step, the display of the vessel into which the catheter is to be positioned is superimposed by current fluoroscopic images (phase C).

Using an existing fill image

The availability of the fill image is shown on the **Road** button and on the footswitch icon on the monitor.

- ❖ Press the **Road** button.
- The *Roadmap* mode is selected.

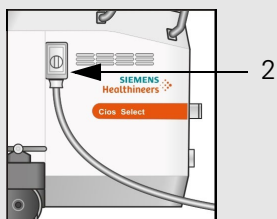
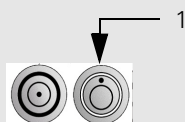
Positioning the catheter (phase C)

- ❖ Release radiation with the hand switch or the **Road** pedal of the footswitch.
- ❖ Position the guidewire or the catheter under fluoroscopic control:
 - The left monitor shows the subtracted Roadmap image plus the catheter.
 - The right monitor shows the fluoro image (native).

Switching off



When you want to turn off the C-arm system only, disconnect it from the monitor trolley.



Switching off

Shutting down the system

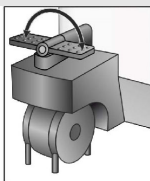
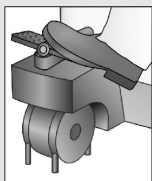
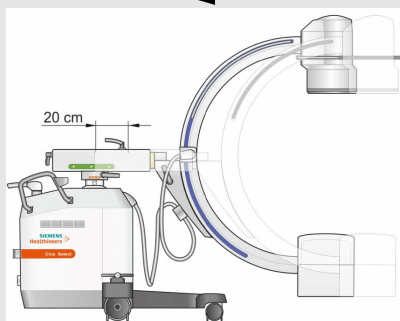
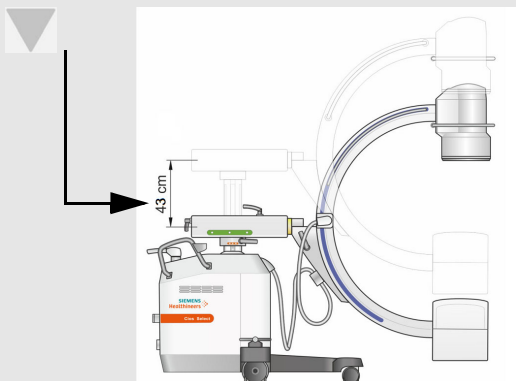
Switching off the system completely

- ❖ Close the current patient with the **Close pat.** button on the control unit.

- ❖ Press the **OFF** button on the monitor trolley (1).
- ❖ Disconnect the monitor trolley power plug from the power outlet.

- ❖ Disconnect the monitor trolley from the C-arm system by unlocking (lever toward the left) and remove the main plug (2).
- ❖ Disconnect the other cables from the C-arm system:
 - ☐ Footswitch
 - ☐ Potential equalization

Switching off



Switching off

Returning to the park position

1. Lower the lifting column

- ❖ Press the down button on the control unit and lower the lifting column to the lowest position.

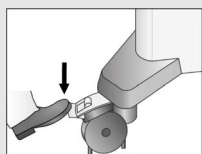
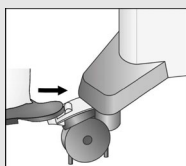
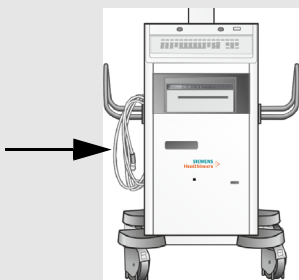
2. Retract the horizontal support arm

- ❖ Release the brake for horizontal movement and move the horizontal support arm all the way back.
- ❖ Press the brake button again to lock it.

3. Engage the brakes

- ❖ Keep the foot pedal on both sides of the system pressed. The foot pedal can be pressed in both directions.
- The basic unit brakes are now locked.

Switching off



On sloping surfaces, make sure that the brakes are locked and the front of the monitor trolley is facing the downward slope. Risk of slipping and tipping!

Transporting the monitor trolley

- ❖ Prior to transport, place the rolled-up cables on the handles of the monitor trolley.

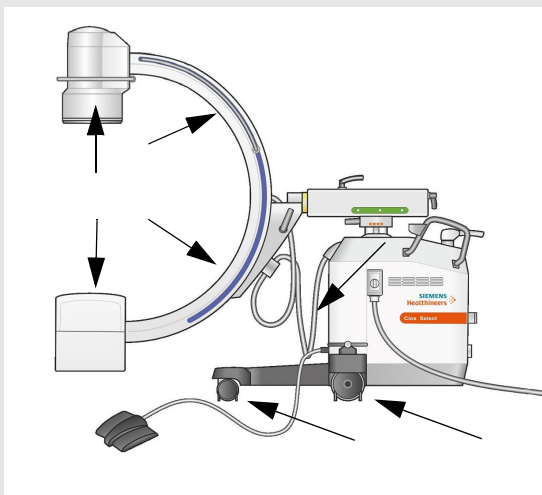
- ❖ Release the main brake at the front of the monitor trolley.

→ The monitor trolley can now be moved.

Park position

- ❖ Using your foot, step down on the main brake until it locks in.

System maintenance



Only use the recommended cleaning agents and disinfectants.

Follow manufacturer's instructions.

Do not allow any cleaning liquids to enter the openings of the system (e.g. air vents, gaps between covers).

Cleaning, disinfection

After you have turned off and disconnected the Cios Select from the main power, you can disinfect or clean the following system components:

- ☐ Basically all components that come in contact with patients
- ☐ X-ray tube assembly housing
- ☐ Image intensifier
- ☐ C-arm
- ☐ Undercarriage (wheels)

Proceed as follows:

- ❖ Clean the system parts with a damp cloth.
- ❖ For moistening, use water or a lukewarm, diluted aqueous solution consisting of water and a household cleaning agent.
- ❖ Do not use scouring cleaning agents or organic solvents or cleaning agents such as benzene, pure alcohol, spot remover etc. because of possible material incompatibility.

To disinfect surfaces, we recommend aqueous solutions of commercially available aldehyde and/or amphotensides-based surface disinfectants such as Tensodur 103, Kohrsolin, Cidex.

Certain substituted phenol-based or chlorine-splitting disinfectants can corrode materials and are therefore not recommended. The same restrictions apply to undiluted solutions with a high alcohol content, e.g. for disinfecting hands.



Alcohol-based cleaning agents are not suitable for screen surfaces and the control units.

Screen surfaces/TFT displays

Screen surfaces should be cleaned at least every two months.

- ❖ Clean the monitor screen with a cotton cloth dampened with water.
- ❖ Remove stubborn stains with a mixture of 2/3 water and 1/3 alcohol.
- ❖ Dry the monitor screen immediately using a soft cotton cloth.
- ❖ Immediately wipe off any splashing from the contrast agent.

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